



Review Test Submission: Assignment Week 10

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Course EBN 122 S2 2017

Test Assignment Week 10

Started 9/26/17 8:28 AM

Submitted 10/15/17 5:19 PM

Due Date 10/16/17 11:59 PM

Status Completed

Attempt 0 out of 30 points

Score

Time 464 hours, 51 minutes

Elapsed

Instructions Read over the document *ClickUP Assignments Instructions*.

The deadline is the time by which the assignment must be SUBMITTED (not just started)

Before sending an enquiring regarding ClickUP question to any-one please verify that your problem is valid ... i.e. if you suspect incorrect marking first check with a fellow student or two.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final correct answer 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- Use SI units unless otherwise specified.
- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use singular)
- Use 'u' to indicate micro.
- Only use pico(p), nano(n), micro(u), milli(m), kilo(k), mega(M), giga(G) and tera(T); do not use centi, deci and deca

If the answer is without units it has no meaning to write milli(m) or kilo(k). Simply write the number 0.1 or 1000

Results All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions

Question 1

0 out of 1 points



Convert the complex number $Z = 90.12 + j59.53$ into polar form, and state its angle in degrees:

Selected Answer: [None Given]

Correct Answer: $33.447 \pm 2\%$

Question 2

0 out of 1 points



What is Θ in:

$$-\sin(13^\circ) = \cos\Theta$$

Give the answer in degrees. Do not indicate the units

Selected Answer: [None Given]

Correct Answer: $103.00 \pm 2\%$

Response Feedback: See paragraph 6.1 of Zekavat.

Question 3

0 out of 1 points



Transform the following sinusoid to phasor form ($x + jy$). Give the imaginary part (y) of your answer below .

$$-58\cos(536t + -22^\circ)$$

Selected Answer: [None Given]

Correct Answer: $21.727 \pm 2\%$

Question 4

0 out of 1 points



The addition of two signals, $x_1(t) = A\cos(\omega t + B^\circ)$ and $x_2(t) = C\cos(\omega t + D^\circ)$ will result in a signal $x(t) = E\cos(Kt + F)$.

Determine the value of E , given: $A = 4$, $C = 11$, $\omega = 752$ rad/s, $B = -114^\circ$ and $D = 83^\circ$. Do not indicate the units

Selected Answer: [None Given]

Correct Answer: $7.269 \pm 2\%$

Question 5

0 out of 1 points



The addition of two signals, $x_1(t) = A\cos(\omega t + B^\circ)$ and $x_2(t) = C\cos(\omega t + D^\circ)$ will result in a signal $x(t) = E\cos(Kt + F)$.

Determine the value of F, given: $A = 6$, $C = 9$, $\omega = 182$ rad/s, $B = 87^\circ$ and $D = 50^\circ$. Provide F in degrees. Do not indicate the units

Selected Answer: [None Given]

Correct Answer: $64.672 \pm 2\%$

Question 6

0 out of 1 points



Given the complex numbers $Z_1 = 9 \angle -81^\circ$ and $Z_2 = 11e^{j77^\circ}$, find Z_1 / Z_2 in the form $P \angle Q^\circ$. What is the value of Q?

Selected Answer: [None Given]

Correct Answer: $-158.00 \pm 2\%$

Question 7

0 out of 1 points



Given the complex numbers $Z_1 = 14 \angle -88^\circ$ and $Z_2 = 3e^{j55^\circ}$, find Z_1 / Z_2 in the form $P \angle Q^\circ$. What is the value of P?

Selected Answer: [None Given]

Correct Answer: $4.67 \pm 2\%$

Question 8

0 out of 1 points



Given the complex numbers $Z_1 = 66 \angle 73^\circ$ and $Z_2 = 77e^{j87.54^\circ}$, find $Z_1 + Z_2$ in the form $P + jQ$. What is the value of P.

Selected Answer: [None Given]

Correct Answer: $22.60 \pm 2\%$

Question 9

0 out of 1 points



Given the complex numbers $Z_1 = 18.31 \angle 93.22$ and $Z_2 = 26.27e^{j92.46}$, find complex conjugate of Z_1 in exponential form Pe^{jQ} , where the value of P is:

Selected Answer: [None Given]

Correct Answer: $18.31 \pm 5\%$

Question 10

0 out of 1 points



Given the complex numbers $Z_1 = 14 \angle 62^\circ$ and $Z_2 = 12e^{j-110^\circ}$, find $Z_1 * Z_2$ in the form $P \angle Q^\circ$.

What is the value of P?

Selected Answer: [None Given]

Correct Answer: 168.00 ± 2%

Question 11

0 out of 1 points



Given the complex numbers $Z_1 = 16 \angle -48^\circ$ and $Z_2 = 7e^{j-44^\circ}$, find $Z_1 * Z_2$ in the form $P \angle Q^\circ$.

What is the value of Q? Do not indicate the units

Selected Answer: [None Given]

Correct Answer: -92.00 ± 2%

Question 12

0 out of 2 points



By how many degrees is the voltage leading.

$$\bar{V}_1 = 8 \angle \theta$$

$$i_1(t) = 10 \sin(\omega t - \phi^\circ)$$

$$\theta = 82^\circ \text{ and } \phi = 49^\circ.$$

The answer must be in degrees. No need to indicate the unit.

Selected Answer: [None Given]

Correct Answer: 221.000 ± 2%

Response Feedback: Remember to convert the sin to a cos.

Question 13

0 out of 1 points



$$x(t) = 3\cos(200\pi t + 45) + 4\sin(100\pi t + 45)$$

in phasor notation is:

Selected Answer: [None Given]

Answers: a. $7 \angle 90^\circ$

a.

b. $5 \angle 90^\circ$

b.

c. $5 \angle 0^\circ$

c.

$7 \angle 45^\circ$
d.

☒ e. None of the above.

Response
Feedback:

No, you can not combine two functions of different frequencies into one phasor.

Question 14

0 out of 1 points



For a sinusoid $v(t) = 49\cos(785t+157^\circ)$ V, determine its phase. Do not indicate the unit

Selected Answer: ☒ [None Given]

Correct Answer: ☒ $157.00 \pm 2\%$

Question 15

0 out of 1 points



For a sinusoid $v(t) = 18.13\cos(531.05t+1.4)$ V, determine its amplitude. Enter the appropriate unit.

Selected Answer: ☒ [None Given]

Correct Answer: ☒ $18.13 \pm 1\%$ (V)

Question 16

0 out of 2 points



Simplify $v(t)$ and write as a single sinusoid, $v(t)=A\cos(\omega t+\theta^\circ)$, with $A \geq 0$ and $-180 < \theta \leq 180$

$$v(t) = 6\cos(\omega t - 49^\circ) + 3\sin(\omega t + 3^\circ) \text{ V}$$

What is the value of θ . No need to indicate the units.

Selected Answer: ☒ [None Given]

Correct Answer: ☒ $-61.452 \pm 2\%$

Question 17

0 out of 2 points



Write in rectangular form

$$\bar{I} = \frac{(3 + j4) + 11 \angle 90^\circ}{(2 \angle 45^\circ)^2 + (3 - j4)}$$

Provide the imaginary part. Do not provide units.

Selected Answer: ☒ [None Given]

Correct Answer: ☒ $5.000 \pm 2\%$

Question 18

0 out of 2 points

Write in rectangular form



$$\bar{I} = \frac{(3 + j4) + 11 \angle 90^\circ}{(2 \angle 45^\circ)^2 + (3 - j4)}$$

Provide the real part. Do not provide units.

Selected Answer:  [None Given]

Correct Answer:  1.000 ± 2%

Question 19

0 out of 2 points



Use phasors to write $v(t)$ as a single cosine expression:

$$v(t) = -4\cos(\omega t - 240^\circ) + 18\sin(\omega t + 120^\circ) \text{ V,}$$

The expression should be in the form:

$$A\cos(\omega t + \phi^\circ) \text{ V}$$

Give the magnitude of A (with $A > 0$ and $-180^\circ < \phi < +180^\circ$)

Selected Answer:  [None Given]

Correct Answer:  18.44 ± 2%

Question 20

0 out of 2 points



Write $v(t)$ in a phasor in rectangular form.

$$v(t) = 4\cos(\omega t + 79^\circ) - 8\sin(\omega t + 36^\circ)$$

The arguments are in radians.

Provide the imaginary part. Do not provide units.

Selected Answer:  [None Given]

Correct Answer:  10.399 ± 2%

Question 21

0 out of 2 points



Simplify the equation below into the polar form. $A \angle B^\circ$ If $X = 9$ give the magnitude A as your answer.

$$\bar{I}_1 = \frac{X(-6 + j4)(5 - j)}{-1 + j\sqrt{3}}$$

Selected Answer:  [None Given]

Correct Answer:  165.46 ± 2%

Question 22

0 out of 2 points



Determine $\mathbf{V}_2$ in polar notation.

$$\begin{bmatrix} 2 + j4 & -j2 \\ -j2 & 2 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \angle 90^\circ \end{bmatrix}$$

Provide the magnitude. Do not provide the units.

Selected Answer:  [None Given]

Correct Answer:  1.58 ± 2%

Sunday, October 15, 2017 5:20:04 PM SAST

← OK